

End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

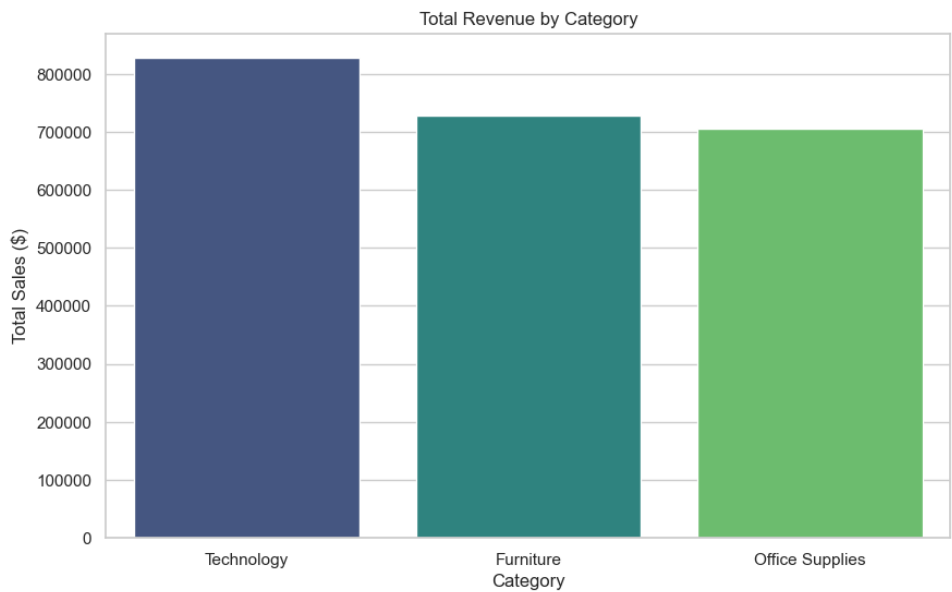
This project analyzes historical sales data to extract insights, forecast future sales, detect anomalies, and segment products for inventory optimization.

1. Exploratory Data Analysis & Insights

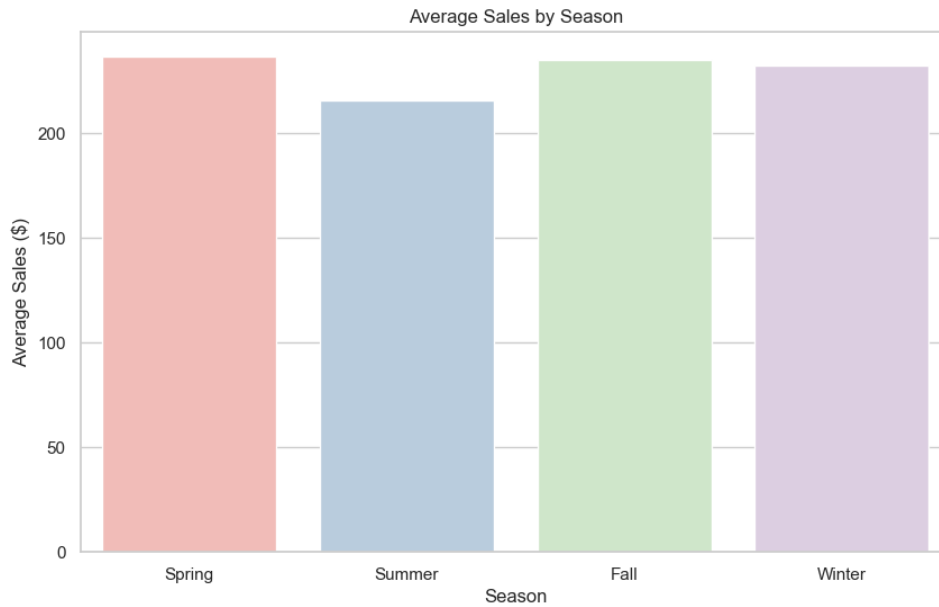
Key findings from historical sales:

- Technology leads in revenue, followed closely by Furniture.
- The Central region exhibits higher variance in sales consistency.
- Fall and Winter seasons typically drive higher sales.

Revenue by Category



Average Sales by Season

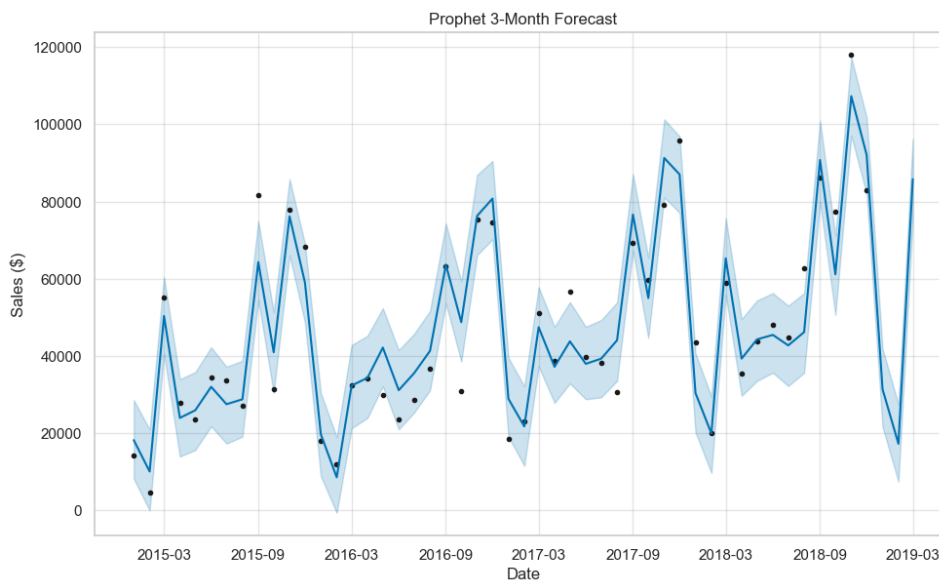


2. Forecasting Models

We tested SARIMA, Facebook Prophet, and XGBoost models on monthly aggregated sales data. The model metrics are summarized below:

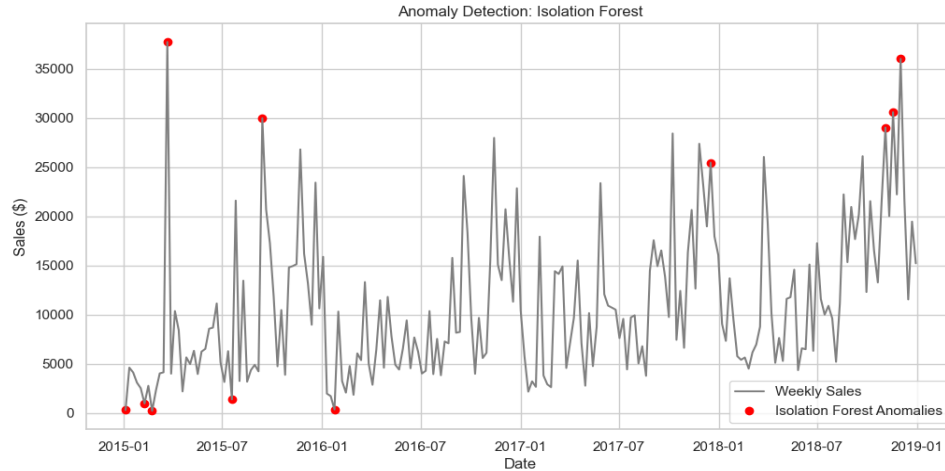
Model	MAE	RMSE	MAPE (%)
SARIMA	14862.39	17299.71	17.89
Prophet	17421.14	20294.61	22.67
XGBoost	21741.78	25076.08	26.22

Prophet 3-Month Forecast



3. Anomaly Detection

Using Isolation Forest and Rolling Z-Score techniques, we identified specific weeks where sales significantly deviated from the norm, indicating potential supply chain disruptions or exceptional demand events.



4. Product Demand Segmentation

Sub-categories were grouped into 4 distinct clusters to guide inventory strategy:

1. Cash Cows (High Volume): Automate reordering.
2. High Growth, High Value: Prioritize stock.
3. Volatile / Seasonal: Use JIT inventory.
4. Steady & Low Value: Optimize storage.

